

Paper II — Centrifugal Vapour-Swirl Contact Tray (CVS): A Deck-Pool-Fed Swirl-Tube Stage Exceeding the Gravity System Limit

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Invention Disclosure / Provisional Specification Draft — HyTray Series HT-02

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Abstract

A vapour–liquid contacting tray is disclosed in which the active area comprises an array of short axial swirl tubes, each fed tangentially by 4–8 vapour slots generating a 30–60 g centrifugal field, and fed with liquid from the tray deck pool through 3–6 calibrated metering ports per tube. Mass transfer occurs in a high-shear swirling annular film and droplet cloud inside each tube; entrainment is removed mechanically by a vaned crown separator at the tube exit which slings liquid into a segregated collection annulus draining directly to the outlet-weir region, bypassing the active deck. Because phase disengagement is centrifugal rather than gravitational, the tray’s useful capacity is decoupled from tray spacing and from the Souders–Brown settling limit: the predicted capacity factor at flood is $C_{sb} = 0.14\text{--}0.18\text{ m/s}$, i.e. 50–80% above conventional cross-flow trays and 30–50% above current high-capacity decks, at tray spacings as low as 300 mm. The disclosed inventive matter resides in the deck-pool-fed multi-port liquid metering, the slot/swirl-number band that fixes the annular-film regime, and the crown-and-annulus geometry that prevents re-entrainment — not in the bare concept of swirl contacting, which is prior art.

Status note: performance values are model predictions (swirl-separator efficiency from droplet-trajectory analysis in a solid-body vortex; film mass transfer from penetration theory at the imposed shear). They require CFD and hot-rig validation. Shell ConSep™/swirl-tube art and cyclonic-contactor patents are dense prior art; a professional search is mandatory before filing.

1 Field of the Invention

Vapour–liquid contacting internals for distillation and absorption columns operating at or beyond the gravity-settling capacity limit of conventional trays, particularly for capacity revamps within an existing shell.

2 Background and Prior Art

The capacity of every conventional tray is bounded by the Souders–Brown criterion: above a superficial vapour velocity $U_s = C_{sb}\sqrt{(\rho_L - \rho_V)/\rho_V}$, droplets generated at the froth surface can no longer settle against the rising vapour within the available tray spacing, and the column floods by entrainment. High-capacity decks (Koch-Glitsch SUPERFRAC®, Sulzer VGPlus™, UFM™) recover area from the downcomers and shape the froth, but remain gravity-limited: the “system limit” (ultimate capacity) stands at roughly $C_{sb} \approx 0.12\text{ m/s}$ for ordinary systems.

Swirl-tube stages (Shell ConSep™ class) demonstrated that centrifugal de-entrainment can exceed this limit, but known designs feed liquid to the contactor from dedicated downcomer hardware and return separated liquid onto the active deck, where part of it is re-entrained; their contactor internals also leave the annular-film regime uncontrolled at turndown. The present invention addresses both deficiencies.

3 Objects of the Invention

1. To provide a tray stage whose flood point is set by mechanical separation capacity, not gravity settling, with $C_{sb} \geq 0.14$ m/s.
2. To make capacity substantially independent of tray spacing down to 300 mm.
3. To meter liquid into each contactor directly from the deck pool, eliminating dedicated feed hardware and ensuring per-tube L/V uniformity.
4. To return separated liquid to the outlet-weir region through a segregated annulus, preventing re-entrainment.
5. To hold point efficiency $E_{OG} \geq 0.8$ across a 3:1 operating window by fixing the annular-film Reynolds and swirl-number bands.

4 Summary of the Invention

Each contactor cell comprises a vertical tube (20) of 80–150 mm inside diameter and height 1.2–2.0 diameters, penetrating the deck. Vapour enters through 4–8 tangential slots (22) at 18–35 m/s, establishing a vortex of swirl number 1.5–3.0 and wall acceleration 30–60 g around the vortex core (24). Liquid enters each tube from the surrounding deck pool through 3–6 calibrated ports (26) of 8–14 mm diameter located in the tube wall below the slots; port sizing fixes the per-tube liquid load at 1.5–4.0 m³/h and maintains the annular film (32) in a claimed film-Reynolds band ($Re_f = 4\Gamma/\mu_L = 1,500$ –8,000). At the tube exit a crown separator (28) of 8–16 vanes set at 35–55° co-rotating pitch intercepts the swirling two-phase flow, slings liquid outward into a collection annulus (30) which drains via sealed conduits directly to the outlet-weir region of the same tray, while dry vapour passes to the tray above.

5 Brief Description of the Drawings

Figure 1: (a) plan section of one swirl tube showing tangential slots, liquid metering ports and vortex core; (b) elevation showing tube, annular film, crown separator and collection annulus. Figure 2: (a) predicted capacity factor versus tray spacing against conventional and high-capacity decks; (b) predicted entrainment suppression versus percent of flood.

6 Detailed Description

Vapour path and swirl generation. Slot width and count are selected so that slot exit velocity stays in the 18–35 m/s band across the operating window; below 18 m/s the vortex weakens (<30 g) and separation efficiency falls, above 35 m/s erosion and ΔP penalties dominate. Swirl number $S = \dot{G}_\theta/(\dot{G}_x R)$ is held between 1.5 and 3.0; in this band the flow is a stable Rankine-type vortex without central reverse-flow breakdown reaching the liquid ports.

Liquid metering from the deck pool. The deck between tubes carries a clear-liquid pool of 40–80 mm maintained by the outlet weir. Each port behaves as a submerged orifice: $Q_p = C_d A_p \sqrt{2gh_{cl}}$ with $C_d \approx 0.62$. Because all tubes share one pool, per-tube liquid load self-equalises — the key advantage over fed-from-above designs in which maldistribution between contactors directly

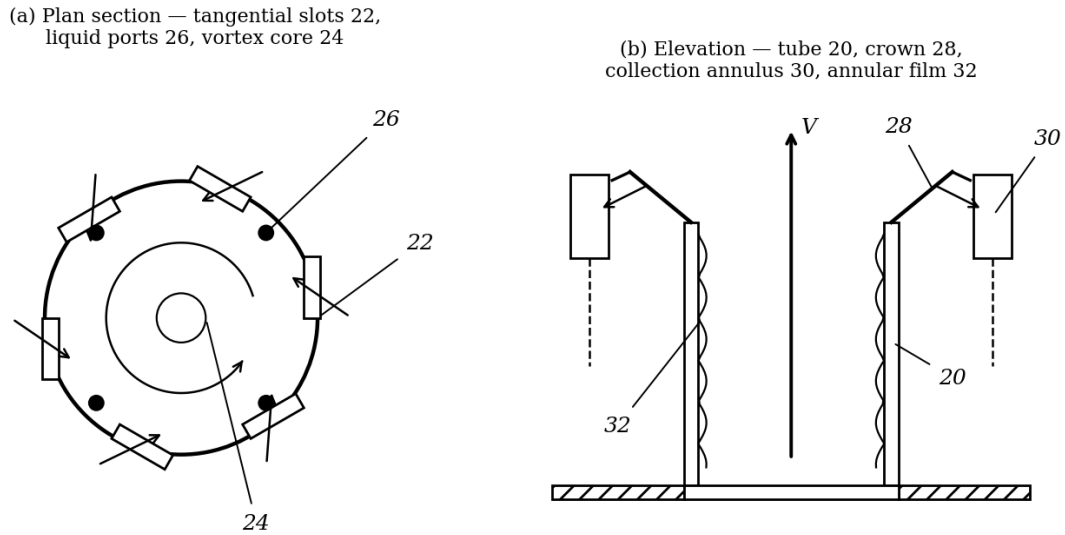


Figure 1: Swirl contactor cell: tube (20), tangential slots (22), vortex core (24), liquid metering ports (26), crown separator (28), collection annulus (30), annular film (32).

destroys efficiency. Port elevation is below the slots so entering liquid is immediately sheared into the rotating film rather than falling through the vortex core.

Mass transfer. The swirling film experiences interfacial shear an order of magnitude above gravity-flow trays; surface renewal scales with u_τ , and the droplet cloud stripped from the film crest provides additional area. Penetration-theory estimates give $E_{OG} = 0.80\text{--}0.92$ for one tube pass at design; tube height/diameter ratio 1.2–2.0 sets residence time.

Crown separation and re-entrainment control. Droplet cut size for a vortex of acceleration a and radial travel R within residence t : $d_{50} \propto \sqrt{18\mu_V R / (a \Delta \rho t)}$; at 45 g and 110 mm tube this is below $40 \mu\text{m}$ — finer than the gravity-settled cut at any practical spacing. The crown vanes turn the separated film into the annulus (30); the annulus drains below the froth surface near the outlet weir, so collected liquid never re-crosses active bubbling area. Sealing legs prevent vapour bypass up the drain conduits.

Layout and hydraulics. Tubes occupy 25–40% of column area on a triangular pitch of 1.4–1.7 tube diameters. Tray ΔP is 8–14 mbar (slots + film + crown), deliberately traded against the 50–80% capacity gain; per theoretical stage the penalty versus a high-capacity tray is $\leq 30\%$ because fewer, shorter stages are needed at 300–350 mm spacing. Turndown of the basic cell is $\sim 3:1$; a dual-slot-row variant (upper row blanked by a static weight-loaded band) extends this to 5:1 and is reserved as a divisional.

7 Design Parameter Envelope (claim-band table)

8 Predicted Performance

9 Claims (draft)

Claim 1. A vapour–liquid contacting tray comprising a deck supporting a liquid pool and a plurality of vertical swirl tubes penetrating said deck, each tube having a plurality of tangential vapour-inlet slots and a plurality of liquid-metering ports opening from said pool into the tube below said slots, the ports being sized such that the liquid film established on the inner tube wall

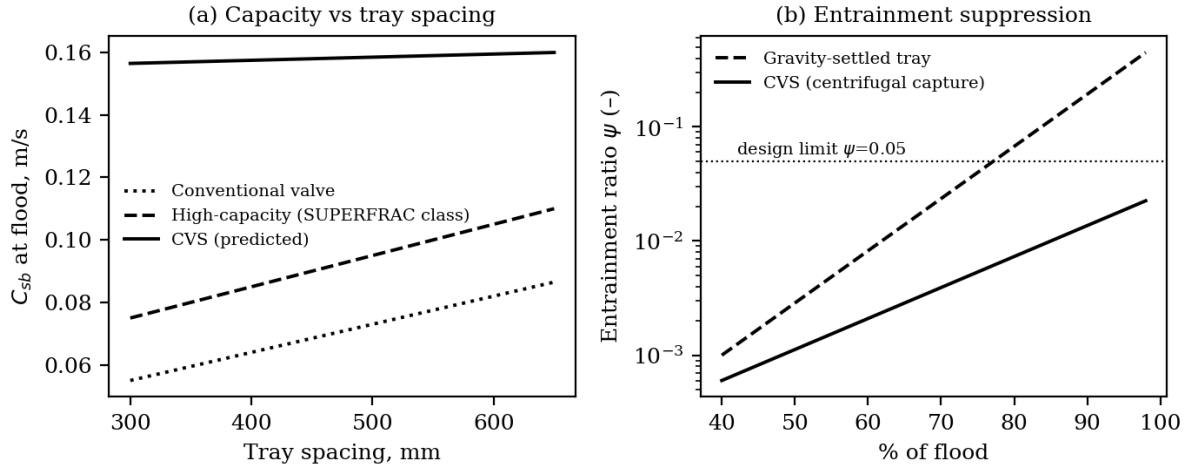


Figure 2: Predicted performance (model-based, pending validation): (a) capacity decoupled from spacing; (b) entrainment held below design limit to >95% of flood.

Parameter	Claimed band	Preferred
Tube ID	80–150 mm	110 mm
Tube height / ID	1.2–2.0	1.6
Tangential slots per tube	4–8	6
Slot exit velocity	18–35 m/s	25 m/s
Swirl number S	1.5–3.0	2.2
Wall acceleration	30–60 g	45 g
Liquid ports per tube	3–6, $\varnothing$ 8–14 mm	4 $\times$ $\varnothing$ 10
Per-tube liquid load	1.5–4.0 m ³ /h	2.5
Film Reynolds $4\Gamma/\mu_L$	1,500–8,000	4,000
Crown vanes / pitch angle	8–16 / 35–55°	12 / 45°
Tube area fraction	25–40 %	32 %
C_{sb} at flood	0.14–0.18 m/s	0.16
Tray spacing	300–450 mm	350 mm
ΔP per tray	8–14 mbar	11

Table 1: Claim-band table; ranges are dependent-claim fallback positions.

has a film Reynolds number $4\Gamma/\mu_L$ of 1,500 to 8,000 throughout a design operating window.

Claim 2. The tray of claim 1, wherein each tube further comprises at its upper end a crown separator of 8–16 vanes pitched co-rotationally at 35–55°, discharging separated liquid into a collection annulus.

Claim 3. The tray of claim 2, wherein the collection annulus drains through sealed conduits to the outlet-weir region of the same tray without traversing active bubbling area.

Claim 4. The tray of claim 1, wherein the slots produce a swirl number of 1.5–3.0 and a wall centripetal acceleration of 30–60 g at design vapour load.

Claim 5. The tray of claim 1, wherein the tubes have an inside diameter of 80–150 mm, a height of 1.2–2.0 diameters, and occupy 25–40% of the column cross-section.

Claim 6. The tray of claim 1, operating at a capacity factor C_{sb} of at least 0.14 m/s at flood at a tray spacing of 450 mm or less.

Claim 7. A method of revamping a distillation column comprising replacing existing cross-flow trays with trays according to any preceding claim at a tray spacing of 300–450 mm, thereby

Metric	Conventional valve	High-capacity deck	CVS (predicted)
C_{sb} at flood, 450 mm TS	0.085	0.105	0.16
Capacity at 300 mm TS (rel.)	0.62	0.78	0.97
E_{OG} at design	0.72	0.80	0.86
Entrainment ψ at 90% flood	0.12	0.06	<0.02
ΔP / theoretical stage	8 mbar	6.5 mbar	12 mbar
Shell capacity gain in revamp	base	+30 %	+55–70 %

Table 2: Standard hydrocarbon system, SF = 1.0; CVS values are correlation/trajectory-model predictions.

increasing column capacity by at least 50% within the existing shell.

Claim 8. A column containing trays according to any preceding claim.

10 Industrial Applicability

Shell-limited revamps: demethanisers, C3 splitters, FCC main fractionator top sections, quench towers, offshore columns where diameter and height carry extreme cost premiums.

Prior-art neighbours to clear: Shell swirl-tube/ConSep patents and continuations, cyclonic tray art in CN/RU literature, axial-cyclone scrubber internals. The asserted inventive step is the deck-pool multi-port metering with claimed film-regime band, plus the segregated crown-to-weir liquid return.